Unit III: Continuous Distributions

Uniform, Exponential, Gamma, Beta, Normal, Transformations, Probability Integral Transformation

STA201 – Theory of Probability and Statistics

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1 Motivation: Why Continuous Distributions?

In Unit II, we studied discrete random variables, which take values that can be listed: $0, 1, 2, \ldots$ or $x_1, x_2, \ldots$. However, many engineering quantities are not discrete — they vary continuously:

- The lifetime of a component (in hours) can be any positive real number.
- The voltage across a resistor at any instant is a real number.
- The time between successive arrivals at a server is continuous.
- The proportion of defective items in a batch takes any value in $[0, 1]$.
- The tensile strength of a steel rod is measured on a continuum.

For such quantities, the probability of any single exact value is zero, and we must work with **intervals** rather than points. This leads to the concept of a continuous random variable and its **probability density function (PDF)**.

The major continuous distributions we will study in this unit are:

1. Continuous Uniform
2. Exponential
3. Gamma
4. Beta
5. Normal

We will then study two powerful tools: **transformation of random variables** (how to find the distribution of a function of a random variable) and the **probability integral transformation** (which connects every continuous distribution to the uniform distribution).

2 A Short Note on the Gamma Function

Before we introduce the gamma distribution, we need to define the **gamma function**, because it appears in the normalisation constants of several continuous distributions (including the gamma and beta distributions). Students who understand the gamma function will find the PDFs of these distributions much easier to work with.

2.1 Definition

For any real number $\alpha > 0$, the gamma function is defined by the integral

$$\boxed{\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, dx.}$$

For $\alpha \leq 0$, the integral diverges, so the gamma function is defined only for positive arguments.

2.2 Properties

1. Recursion formula. Integrating by parts:

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, dx.$$

Let $u = x^{\alpha-1}$, $dv = e^{-x}\, dx$, so $du = (\alpha-1)x^{\alpha-2}\, dx$, $v = -e^{-x}$:

$$\Gamma(\alpha) = \left[-x^{\alpha-1}e^{-x}\right]_0^\infty + (\alpha-1)\int_0^\infty x^{\alpha-2} e^{-x}\, dx.$$

The boundary term vanishes (for $\alpha > 1$), so

$$\boxed{\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1).}$$

2. Factorials. Repeatedly applying the recursion:

$$\Gamma(n) = (n-1)(n-2)\cdots 1 \cdot \Gamma(1) = (n-1)!.$$

Since $\Gamma(1) = \int_0^\infty e^{-x}\,dx = 1$, we have

$$\boxed{\Gamma(n) = (n-1)! \quad \text{for integer } n \geq 1.}$$

3. Half-integer values. The value at $\alpha = 1/2$ is

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty x^{-1/2} e^{-x}\,dx = \sqrt{\pi}.$$

(This is the Gaussian integral in disguise.) Combining with the recursion:

$$\Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}, \qquad \Gamma\left(\frac{5}{2}\right) = \frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi} = \frac{3\sqrt{\pi}}{4}.$$

In general, for a positive integer n:

$$\Gamma\left(n + \frac{1}{2}\right) = \frac{(2n)!}{4^n\, n!}\sqrt{\pi}.$$

4. A useful integral. The substitution $u = x/\beta$ (for $\beta > 0$) gives

$$\int_0^\infty x^{\alpha-1} e^{-x/\beta}\,dx = \beta^\alpha \Gamma(\alpha).$$

This identity appears repeatedly in the normalisation of the gamma distribution.

2.3 Examples

Solved Example Evaluating Gamma Function Values

Compute $\Gamma(5)$, $\Gamma(1)$, $\Gamma(1/2)$, and $\Gamma(7/2)$.

Solution:

- $\Gamma(5) = 4! = 24$.
- $\Gamma(1) = 0! = 1$.
- $\Gamma(1/2) = \sqrt{\pi} \approx 1.7725$.
- $\Gamma(7/2) = \frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi} = \frac{15\sqrt{\pi}}{8} \approx 3.3234$.

Solved Example Using the Recursion Formula

If $\Gamma(\alpha) = 6$ and $\alpha > 0$, find α assuming it is an integer.

Solution: If $\alpha = n$ is an integer, $\Gamma(n) = (n-1)! = 6$. The only integer factorial equal to 6 is $3! = 6$, so $n - 1 = 3$, i.e., $\alpha = 4$.

Solved Example Evaluating an Integral with Gamma Function

Evaluate $\displaystyle\int_0^\infty x^3 e^{-2x}\,dx$.

Solution: Write $x^3e^{-2x} = x^{(4-1)}e^{-x/(1/2)}$, which has the form $x^{\alpha-1}e^{-x/\beta}$ with $\alpha = 4$ and $\beta = 1/2$:

$$\int_0^\infty x^3 e^{-2x}\,dx = \beta^\alpha \Gamma(\alpha) = \left(\frac{1}{2}\right)^4 \Gamma(4) = \frac{1}{16}\cdot 6 = \frac{3}{8}.$$

These values and identities will be used repeatedly in the sections on the gamma and beta distributions.

3 Continuous Uniform Distribution

3.1 Motivation

The continuous uniform distribution models a random variable whose value is equally likely anywhere in an interval. Examples: rounding errors, random number generators, unknown waiting times, random angles.

3.2 Definition

$X \sim U(a,b)$ if

$$\boxed{f(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}}$$

CDF: $F(x) = \dfrac{x-a}{b-a}$ for $a \le x \le b$.

3.3 Moments

$$E[X] = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$

$$M_X(t) = \frac{e^{bt} - e^{at}}{t(b-a)}, \quad t \neq 0.$$

3.4 Examples

Solved Example Rounding Error of a Digital Display

A digital thermometer rounds the measured temperature to the nearest integer. The true temperature is uniformly distributed over $[20.5, 21.5]$. The rounding error X is uniform on $[-0.5, 0.5]$. Find $E[X]$, $\mathrm{Var}(X)$, and $P(|X| \le 0.25)$.

Solution: $a = -0.5, b = 0.5$.

$$E[X] = 0, \quad \mathrm{Var}(X) = \frac{1^2}{12} \approx 0.0833, \quad P(|X| \le 0.25) = 0.5.$$

Solved Example Waiting for a Bus

Buses arrive every 15 minutes. If you arrive at a random time, the waiting time $W \sim U(0, 15)$. Find $P(W > 10)$, $E[W]$, and $\mathrm{Var}(W)$.

Solution:

$$P(W > 10) = \frac{5}{15} = \frac{1}{3}, \quad E[W] = 7.5, \quad \mathrm{Var}(W) = \frac{225}{12} = 18.75.$$

Solved Example Random Number Generator

A random number generator produces $U \sim U(0, 1)$. Find $P(0.2 < U < 0.6)$ and the MGF.

Solution:

$$P(0.2 < U < 0.6) = 0.4, \quad M_U(t) = \frac{e^t - 1}{t}.$$

Solved Example Quality Control with Uniform Tolerance

A machined dimension is uniformly distributed on $[48, 52]$ mm. Specifications allow dimensions between 49 and 51 mm. Find the probability a part is within spec, and the expected dimension.

Solution: $X \sim U(48, 52)$.

$$P(49 \le X \le 51) = \frac{51 - 49}{52 - 48} = \frac{2}{4} = 0.5, \quad E[X] = \frac{48 + 52}{2} = 50.$$

Only half the parts meet spec.

Solved Example Random Phase Angle

In a communication system, the phase angle Θ of a carrier wave is uniformly distributed on $[0, 2\pi)$. Find $E[\Theta]$ and $P(\Theta < \pi/2)$.

Solution:

$$E[\Theta] = \frac{0 + 2\pi}{2} = \pi, \quad P(\Theta < \pi/2) = \frac{\pi/2}{2\pi} = \frac{1}{4}.$$

Solved Example Two-Dimensional Uniform

A point (X, Y) is chosen uniformly in the unit square $[0, 1]^2$. Find the probability that the point lies inside the circle $x^2 + y^2 \le 1$.

Solution: The joint PDF is $f(x, y) = 1$ on $[0, 1]^2$.

$$P(X^2 + Y^2 \le 1) = \text{area of quarter circle} = \frac{\pi/4}{1} = \frac{\pi}{4}.$$

This is the theoretical basis of the Monte Carlo method for estimating π.

4 Exponential Distribution

4.1 Motivation

The exponential distribution models the waiting time between events in a Poisson process, and the lifetime of components with a constant failure rate.

4.2 Definition

$T \sim \text{Exp}(\lambda)$, $\lambda > 0$:

$$\boxed{f(t) = \lambda e^{-\lambda t}, \quad t \geq 0.}$$

CDF: $F(t) = 1 - e^{-\lambda t}$.

4.3 Moments and Memoryless Property

$$E[T] = \frac{1}{\lambda}, \quad \text{Var}(T) = \frac{1}{\lambda^2}, \quad M_T(t) = \frac{\lambda}{\lambda - t} \text{ for } t < \lambda.$$

Memoryless property: $P(T > s + t \mid T > s) = P(T > t)$.

4.4 Examples

Solved Example Component Lifetime

The lifetime (in hours) of a component is exponential with mean 500 hours. Find (a) the rate λ, (b) $P(T \geq 700)$, (c) $P(400 \leq T \leq 700)$.

Solution: $\lambda = 1/500 = 0.002$.

$$P(T \geq 700) = e^{-1.4} \approx 0.2466, \quad P(400 \leq T \leq 700) = e^{-0.8} - e^{-1.4} \approx 0.2027.$$

Solved Example Memoryless Property

For the same component, suppose it has already survived 200 hours. Find $P(T > 500 \mid T > 200)$.

Solution: By the memoryless property,

$$P(T > 500 \mid T > 200) = P(T > 300) = e^{-0.6} \approx 0.5488.$$

Solved Example Poisson Process

Calls arrive at a call centre at 5 per hour (Poisson process). Find (a) the probability the next call arrives within 10 minutes, (b) the expected time until the next call.

Solution: $\lambda = 5/\text{hr}$.

$$P(T \le 1/6) = 1 - e^{-5/6} \approx 0.5654, \quad E[T] = 12 \text{ min}.$$

Solved Example Radioactive Decay

The lifetime of a radioactive atom (in seconds) is $\text{Exp}(\lambda = 0.1)$. Find the median lifetime.

Solution: The median satisfies $F(m) = 0.5$:

$$1 - e^{-0.1m} = 0.5 \implies m = \frac{\ln 2}{0.1} = 10 \ln 2 \approx 6.93 \text{ s}.$$

Note that the mean is 10 s, larger than the median because the distribution is right-skewed.

Solved Example Two Independent Exponentials

Two components have independent exponential lifetimes with rates $\lambda_1 = 0.01$ and $\lambda_2 = 0.02$ (per hour). Find the probability that component 1 outlasts component 2.

Solution: We want $P(T_1 > T_2)$.

$$P(T_1 > T_2) = \int_0^\infty P(T_1 > t) f_{T_2}(t)\, dt = \int_0^\infty e^{-\lambda_1 t} \lambda_2 e^{-\lambda_2 t}\, dt = \frac{\lambda_2}{\lambda_1 + \lambda_2}.$$

$$= \frac{0.02}{0.03} = \frac{2}{3}.$$

Solved Example Minimum of Exponentials

In a parallel system of n independent components with lifetimes $T_i \sim \text{Exp}(\lambda_i)$, the system fails when the *first* component fails. Show that the system lifetime is $\text{Exp}(\lambda_1 + \cdots + \lambda_n)$.

Solution:

$$P(\min T_i > t) = \prod_i P(T_i > t) = \prod_i e^{-\lambda_i t} = e^{-(\sum \lambda_i)t}.$$

So $\min T_i \sim \text{Exp}(\sum \lambda_i)$.

5 Gamma Distribution

5.1 Motivation

The gamma distribution generalises the exponential. It models the time until the r-th event in a Poisson process, the sum of independent exponentials, and many other positive quantities.

5.2 Definition

$X \sim \text{Gamma}(\alpha, \beta)$ with shape $\alpha > 0$ and **scale parameter** $\beta > 0$:

$$\boxed{f(x) = \frac{x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha)\, \beta^{\alpha}}, \quad x > 0.}$$

Note on parameterisation: Some textbooks use the **rate parameter** $\lambda = 1/\beta$, in which case

$$f(x) = \frac{\lambda^{\alpha} x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}.$$

In this note, β is the scale parameter. The formulas below are for the scale parameterisation.

5.3 Verification

Using the identity from the gamma function section, $\int_0^{\infty} x^{\alpha-1} e^{-x/\beta}\, dx = \beta^{\alpha}\Gamma(\alpha)$. Thus

$$\int_0^{\infty} \frac{x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha)\beta^{\alpha}}\, dx = \frac{\beta^{\alpha}\Gamma(\alpha)}{\Gamma(\alpha)\beta^{\alpha}} = 1.$$

5.4 CDF

For integer α,

$$F(x) = 1 - e^{-x/\beta} \sum_{k=0}^{\alpha-1} \frac{(x/\beta)^k}{k!}, \quad x > 0.$$

5.5 Moments

$$E[X] = \alpha\beta, \quad \text{Var}(X) = \alpha\beta^2, \quad M_X(t) = (1 - \beta t)^{-\alpha} \text{ for } t < 1/\beta.$$

5.6 Special Cases

- $\alpha = 1$: $\text{Gamma}(1, \beta) = \text{Exp}(1/\beta)$.
- $\alpha = n/2,\ \beta = 2$: chi-squared with n degrees of freedom.
- $\alpha = r$ integer: time to the r-th event in a Poisson process.

5.7 Examples

Solved Example Time to Third Breakdown

A machine breaks down according to a Poisson process at 0.5 breakdowns per month. Find the probability that the third breakdown occurs within 6 months.

Solution: With scale parameter $\beta = 1/0.5 = 2$ months, $X \sim \text{Gamma}(3, 2)$.

$$F(6) = 1 - e^{-3}\left[1 + 3 + \frac{9}{2}\right] = 1 - e^{-3} \cdot 8.5 \approx 0.5768.$$

Solved Example Mean and Variance of Gamma

If $X \sim \text{Gamma}(4, 3)$, find $E[X]$, $\text{Var}(X)$, and the MGF.

Solution:

$$E[X] = 4 \cdot 3 = 12, \quad \text{Var}(X) = 4 \cdot 9 = 36, \quad M_X(t) = (1 - 3t)^{-4}.$$

Solved Example Sum of Exponentials

If $X_1, X_2, X_3 \sim \text{Exp}(\lambda = 1/2)$ independently, find the distribution of $S = X_1 + X_2 + X_3$.

Solution: Each $X_i = \text{Gamma}(1, 2)$. The sum is $\text{Gamma}(3, 2)$.

Solved Example Chi-Squared as Gamma

Show that a chi-squared distribution with 5 degrees of freedom is a gamma distribution with specific parameters.

Solution: $\chi^2_5 = \text{Gamma}(\alpha = 5/2,\ \beta = 2)$, i.e., shape 2.5 and scale 2.

$$E[\chi^2_5] = \frac{5}{2} \cdot 2 = 5, \quad \text{Var} = \frac{5}{2} \cdot 4 = 10.$$

Solved Example Reliability of a Multi-stage System

A system fails after 4 stages, each with independent exponential lifetimes with mean 100 hours. Find the probability the system survives at least 500 hours.

Solution: The total lifetime is $\text{Gamma}(4, 100)$. Using the CDF:

$$P(T \geq 500) = e^{-5} \sum_{k=0}^{3} \frac{5^k}{k!} = e^{-5} \left(1 + 5 + \frac{25}{2} + \frac{125}{6}\right) \approx 0.006738 \times 39.333 \approx 0.2650.$$

Solved Example Finding the Scale Parameter

The lifetime T of a component is gamma-distributed with shape $\alpha = 3$ and mean 12 months. Find the scale parameter and the probability that $T > 10$ months.

Solution: $E[T] = \alpha\beta = 12 \implies \beta = 4$.

$$P(T > 10) = e^{-10/4} \sum_{k=0}^{2} \frac{(10/4)^k}{k!} = e^{-2.5} \left(1 + 2.5 + 3.125\right) \approx 0.0821 \times 6.625 \approx 0.5437.$$

6 Beta Distribution

6.1 Motivation

The beta distribution is the natural model for a **proportion**, a fraction, or a probability that lies between 0 and 1. It appears in three contexts:

1. **Direct modelling of proportions:** the fraction of defective items in a batch, the fraction of time a machine is operational, the fraction of a project completed by a deadline, the fraction of impurities in a chemical mixture.
2. **Bayesian statistics:** the beta distribution is the *conjugate prior* for the binomial parameter. If a prior $\text{Beta}(\alpha, \beta)$ is placed on the unknown success probability p, and we observe s successes and f failures, the posterior is $\text{Beta}(\alpha + s, \beta + f)$.
3. **Order statistics:** if $U_{(1)} \le U_{(2)} \le \cdots \le U_{(n)}$ are the order statistics of n i.i.d. $U(0,1)$ random variables, then the k-th order statistic $U_{(k)} \sim \text{Beta}(k, n-k+1)$. In particular, the minimum is $\text{Beta}(1, n)$ and the maximum is $\text{Beta}(n, 1)$.

6.2 Definition

$X \sim \text{Beta}(\alpha, \beta)$ with $\alpha > 0$, $\beta > 0$ if

$$\boxed{f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, \quad 0 < x < 1,}$$

where the **beta function** is

$$B(\alpha,\beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\,dx = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

The beta function is symmetric: $B(\alpha,\beta) = B(\beta,\alpha)$.

6.3 Verification

By construction,

$$\int_0^1 \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}\,dx = \frac{B(\alpha,\beta)}{B(\alpha,\beta)} = 1.$$

Also $f(x) \ge 0$ on $(0,1)$.

6.4 Shape of the PDF

The shape of the beta density depends on α and β:

- $\alpha = \beta = 1$: uniform on $(0,1)$.
- $\alpha = \beta > 1$: symmetric, unimodal, peaked at $x = 1/2$.
- $\alpha > 1$, $\beta > 1$, $\alpha \ne \beta$: unimodal, skewed toward the smaller parameter.
- $\alpha < 1$, $\beta < 1$: U-shaped, with singularities at 0 and 1 (arcsine distribution when $\alpha = \beta = 1/2$).
- $\alpha < 1$, $\beta \ge 1$: monotone decreasing, with a singularity at 0.
- $\alpha \ge 1$, $\beta < 1$: monotone increasing, with a singularity at 1.

6.5 Mean, Variance, and Mode

$$E[X] = \frac{\alpha}{\alpha+\beta}.$$

$$\text{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

$$\text{Mode} = \frac{\alpha-1}{\alpha+\beta-2}, \quad \alpha > 1,\ \beta > 1.$$

The **higher moments** are

$$E[X^k] = \frac{B(\alpha+k,\beta)}{B(\alpha,\beta)} = \frac{\Gamma(\alpha+k)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+\beta+k)}.$$

6.6 Symmetry Property

If $X \sim \text{Beta}(\alpha,\beta)$, then

$$1 - X \sim \text{Beta}(\beta,\alpha).$$

This implies that the mean of $1-X$ is $\beta/(\alpha+\beta)$, confirming the symmetry of the family. In particular, when $\alpha = \beta$, the distribution is symmetric about $1/2$, and therefore the mean, median, and mode are all $1/2$.

6.7 Special Cases

- $\text{Beta}(1,1) = U(0,1)$.
- $\text{Beta}(1/2,1/2)$: arcsine distribution, PDF $f(x) = \frac{1}{\pi\sqrt{x(1-x)}}$.
- $\text{Beta}(2,2)$: parabolic density $f(x) = 6x(1-x)$, symmetric.
- $\text{Beta}(3,1)$: $f(x) = 3x^2$, monotone increasing.
- $\text{Beta}(1,3)$: $f(x) = 3(1-x)^2$, monotone decreasing.

6.8 Examples

Solved Example Identifying a Beta Density

A random variable has PDF $f(x) = 12x^2(1-x)$, $0 < x < 1$. Verify that it is a beta density and identify the parameters.

Solution: Compare with the beta form $x^{\alpha-1}(1-x)^{\beta-1}/B(\alpha,\beta)$. We have $\alpha - 1 = 2 \implies \alpha = 3$ and $\beta - 1 = 1 \implies \beta = 2$. Check the constant:

$$B(3,2) = \frac{\Gamma(3)\Gamma(2)}{\Gamma(5)} = \frac{2\cdot 1}{24} = \frac{1}{12}.$$

Thus $f(x) = x^2(1-x)/B(3,2) = 12x^2(1-x)$, and $X \sim \text{Beta}(3,2)$.

Solved Example Mean, Variance, and Mode of Beta(3,2)

For $X \sim \text{Beta}(3,2)$, find $E[X]$, $\text{Var}(X)$, and the mode.

Solution:

$$E[X] = \frac{3}{3+2} = 0.6, \quad \text{Var}(X) = \frac{3\cdot 2}{5^2\cdot 6} = \frac{6}{150} = 0.04, \quad \text{Mode} = \frac{3-1}{3+2-2} = \frac{2}{3} \approx 0.667.$$

The mode exceeds the mean ($0.667 > 0.6$), consistent with the right skew of this beta density.

Solved Example Probability for Beta(3,2)

For $X \sim \text{Beta}(3,2)$, find $P(X > 0.5)$.

Solution: The PDF is $f(x) = 12x^2(1-x)$.

$$P(X > 0.5) = \int_{0.5}^{1} 12x^2(1-x)\,dx = 12\int_{0.5}^{1}(x^2 - x^3)\,dx = 12\left[\frac{x^3}{3} - \frac{x^4}{4}\right]_{0.5}^{1}.$$

At $x = 1$: $\frac{1}{3} - \frac{1}{4} = \frac{1}{12}$. At $x = 0.5$: $\frac{1}{24} - \frac{1}{64} = \frac{8-3}{192} = \frac{5}{192}$. Thus

$$P(X > 0.5) = 12\left(\frac{1}{12} - \frac{5}{192}\right) = 12 \cdot \frac{16-5}{192} = 12 \cdot \frac{11}{192} = \frac{11}{16} = 0.6875.$$

Solved Example Proportion of Defectives

The proportion X of defectives follows $\text{Beta}(2,8)$. Find the mean, variance, and $P(X > 0.4)$.

Solution:

$$E[X] = \frac{2}{10} = 0.2, \quad \text{Var}(X) = \frac{16}{100 \cdot 11} = \frac{16}{1100} \approx 0.01455.$$

Using $B(2,8) = 1/72$, the PDF is $f(x) = 72x(1-x)^7$. Then

$$P(X > 0.4) = 72\int_{0.4}^{1} x(1-x)^7\,dx = 72\left[\frac{u^8}{8} - \frac{u^9}{9}\right]_{0}^{0.6} \approx 0.0706.$$

Solved Example Symmetric Beta: Mode and Probability

For $X \sim \text{Beta}(4,4)$, find the mode and $P(X < 0.5)$.

Solution: Since $\alpha = \beta = 4$, the density is symmetric about $1/2$. The mode is

$$\frac{4-1}{4+4-2} = \frac{3}{6} = 0.5.$$

By symmetry, $P(X < 0.5) = P(X > 0.5) = 0.5$.

Solved Example Uniform as Beta(1,1)

Verify that $\text{Beta}(1,1)$ is $U(0,1)$.

Solution: $B(1,1) = \dfrac{\Gamma(1)\Gamma(1)}{\Gamma(2)} = 1$, so $f(x) = 1$ on $(0,1)$.

Solved Example Finding Parameters from Mean and Variance

If $E[X] = 1/2$ and $\mathrm{Var}(X) = 1/20$, find α and β.

Solution: $E[X] = 1/2 \implies \alpha = \beta$. Setting $\mathrm{Var} = \dfrac{\alpha^2}{(2\alpha)^2(2\alpha+1)} = \dfrac{1}{4(2\alpha+1)} = \dfrac{1}{20}$ gives $2\alpha + 1 = 5$, so $\alpha = \beta = 2$.

Solved Example Arc Sine Distribution

If $X \sim \mathrm{Beta}(1/2, 1/2)$, show that the PDF is $f(x) = \dfrac{1}{\pi\sqrt{x(1-x)}}$ and compute $P(X < 0.25)$.

Solution: First, $B(1/2,1/2) = \dfrac{\Gamma(1/2)\Gamma(1/2)}{\Gamma(1)} = \dfrac{\sqrt{\pi}\cdot\sqrt{\pi}}{1} = \pi$. Thus

$$f(x) = \frac{x^{-1/2}(1-x)^{-1/2}}{\pi} = \frac{1}{\pi\sqrt{x(1-x)}}.$$

Then

$$P(X < 0.25) = \int_0^{1/4} \frac{1}{\pi\sqrt{x(1-x)}}\,dx.$$

Substitute $x = \sin^2\theta$, $dx = 2\sin\theta\cos\theta\,d\theta$, $\sqrt{x(1-x)} = \sin\theta\cos\theta$ (for $0 < \theta < \pi/2$):

$$P(X < 0.25) = \frac{1}{\pi}\int_0^{\pi/6} \frac{2\sin\theta\cos\theta}{\sin\theta\cos\theta}\,d\theta = \frac{1}{\pi}\cdot 2\cdot\frac{\pi}{6} = \frac{1}{3}.$$

Solved Example Machine Uptime

A machine's uptime fraction follows $\mathrm{Beta}(18, 2)$. Find the mean and the probability that the machine is operational more than 85% of the time.

Solution:

$$E[X] = \frac{18}{20} = 0.9, \quad B(18,2) = \frac{17!\,1!}{19!} = \frac{1}{171}.$$

So $f(x) = 171x^{17}(1-x)$. Then

$$P(X > 0.85) = 171\int_{0.85}^{1} x^{17}(1-x)\,dx = 171\left[\frac{x^{18}}{18} - \frac{x^{19}}{19}\right]_{0.85}^{1}.$$

At $x = 1$: $\frac{1}{18} - \frac{1}{19} = \frac{1}{342}$. At $x = 0.85$: $0.85^{18} \approx 0.0536$, $0.85^{19} \approx 0.0456$, so the bracket is

$\approx 0.00298 - 0.00240 = 0.00058$. Therefore

$$P(X > 0.85) \approx 171\,(0.002924 - 0.00058) \approx 0.4008.$$

7 Normal Distribution

7.1 Motivation

The normal distribution is the most important distribution in probability and statistics. Its importance comes from three sources:

1. **Empirical ubiquity:** many natural and engineered quantities are approximately normal — measurement errors, dimensions of manufactured parts, thermal noise, blood pressure, heights.
2. **The Central Limit Theorem:** sums and averages of many independent random variables tend to be normal, regardless of the underlying distribution.
3. **Mathematical tractability:** the normal distribution is closed under linear combinations and has a simple moment generating function, making it the foundation of statistical inference.

7.2 Definition

$X \sim N(\mu, \sigma^2)$ with $\mu \in \mathbb{R}$, $\sigma > 0$:

$$\boxed{f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty.}$$

7.3 Verification

Using the Gaussian integral $\int_{-\infty}^{\infty} e^{-z^2/2}\,dz = \sqrt{2\pi}$, the substitution $z = (x-\mu)/\sigma$ gives

$$\int_{-\infty}^{\infty} f(x)\,dx = \frac{1}{\sigma\sqrt{2\pi}} \cdot \sigma\sqrt{2\pi} = 1.$$

7.4 Properties of the Density Curve

1. **Symmetry:** $f(\mu - x) = f(\mu + x)$. The curve is symmetric about μ.
2. **Maximum:** the peak is at $x = \mu$, with value $f(\mu) = \dfrac{1}{\sigma\sqrt{2\pi}}$.
3. **Inflection points:** at $x = \mu \pm \sigma$, where the second derivative changes sign.
4. **Tails:** $f(x) \to 0$ as $|x| \to \infty$, and the approach is very rapid (exponential decay of the square).
5. **Effect of μ:** changing μ shifts the curve horizontally without changing its shape.
6. **Effect of σ:** increasing σ flattens and widens the curve; decreasing σ makes it taller and narrower. The total area under the curve remains 1.

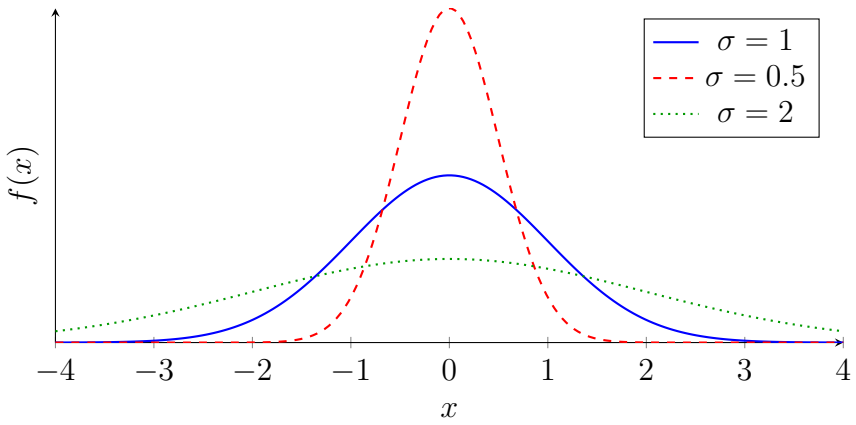


Figure 1: Standard normal densities for three values of σ. All are centred at $\mu = 0$.

7.5 Empirical Rule (68–95–99.7 Rule)

For $X \sim N(\mu, \sigma^2)$,

$$P(\mu - \sigma < X < \mu + \sigma) \approx 0.6827,$$
$$P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.9545,$$
$$P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.9973.$$

This rule is often used in quality control to set control limits.

7.6 Mean, Variance, and MGF

$$E[X] = \mu,$$
$$\mathrm{Var}(X) = \sigma^2,$$
$$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}.$$

All odd central moments are zero (by symmetry), and even central moments satisfy

$$E[(X-\mu)^{2k}] = (2k-1)!!\,\sigma^{2k} = \frac{(2k)!}{2^k k!}\sigma^{2k}.$$

In particular, $E[(X-\mu)^4] = 3\sigma^4$, so the kurtosis of the normal distribution is $\beta_2 = 3$.

7.7 Standard Normal Distribution

The **standard normal** distribution is $Z \sim N(0,1)$:

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad \Phi(z) = \int_{-\infty}^{z} \phi(t)\,dt.$$

The CDF $\Phi(z)$ has no closed form; its values are tabulated or computed numerically.

7.8 Standardisation

If $X \sim N(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim N(0,1).$$

This is a direct consequence of the linear transformation result (see Section 8): if $Y = aX + b$, then $Y \sim N(a\mu + b, a^2\sigma^2)$. Setting $a = 1/\sigma$ and $b = -\mu/\sigma$ gives $Z \sim N(0, 1)$.

Therefore,

$$P(a < X < b) = P\left(\frac{a-\mu}{\sigma} < Z < \frac{b-\mu}{\sigma}\right) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right).$$

7.9 Linear Combinations of Independent Normals

If $X_1, \ldots, X_n$ are independent with $X_i \sim N(\mu_i, \sigma_i^2)$, then

$$Y = \sum_{i=1}^{n} a_i X_i \sim N\left(\sum_{i=1}^{n} a_i\mu_i,\ \sum_{i=1}^{n} a_i^2\sigma_i^2\right).$$

In particular,

$$\sum_{i=1}^{n} X_i \sim N\left(\sum \mu_i, \sum \sigma_i^2\right), \quad \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

This closure property is what makes the normal distribution fundamental in statistical inference.

7.10 Normal Approximation to the Binomial

If $X \sim B(n, p)$ with n large and p not too close to 0 or 1, then

$$\frac{X - np}{\sqrt{np(1-p)}} \approx N(0, 1).$$

The usual rule of thumb: use the approximation when $np \geq 5$ and $n(1-p) \geq 5$. Apply a **continuity correction** when computing $P(a \leq X \leq b)$:

$$P(a \leq X \leq b) \approx P\left(\frac{a - 0.5 - np}{\sqrt{np(1-p)}} \leq Z \leq \frac{b + 0.5 - np}{\sqrt{np(1-p)}}\right).$$

7.11 Examples

Solved Example Basic Standardisation

If $X \sim N(50, 4^2)$, find $P(X < 55)$.

Solution:

$$P(X < 55) = P\left(Z < \frac{55 - 50}{4}\right) = P(Z < 1.25) = \Phi(1.25) \approx 0.8944.$$

Solved Example Probability in an Interval

If $X \sim N(50, 4^2)$, find $P(45 < X < 60)$.

Solution:

$$P(45 < X < 60) = P\left(\frac{45 - 50}{4} < Z < \frac{60 - 50}{4}\right) = P(-1.25 < Z < 2.5).$$

$$= \Phi(2.5) - \Phi(-1.25) = 0.9938 - 0.1056 = 0.8882.$$

Solved Example Symmetric Interval

If $X \sim N(50, 4^2)$, find $P(|X - 50| < 6)$.

Solution:

$$P(|X-50| < 6) = P(44 < X < 56) = P\left(\frac{44-50}{4} < Z < \frac{56-50}{4}\right) = P(-1.5 < Z < 1.5).$$

$$= 2\Phi(1.5) - 1 = 2(0.9332) - 1 = 0.8664.$$

Solved Example Upper Tail

If $X \sim N(100, 15^2)$, find $P(X > 130)$.

Solution:

$$P(X > 130) = P\left(Z > \frac{130-100}{15}\right) = P(Z > 2) = 1 - \Phi(2) = 0.0228.$$

Solved Example Component Dimensions

Shaft diameter $X \sim N(25, 0.08^2)$. Find $P(24.9 < X < 25.1)$.

Solution:

$$P(-1.25 < Z < 1.25) = \Phi(1.25) - \Phi(-1.25) \approx 0.7888.$$

Solved Example Measurement Error

Measurement error $E \sim N(0, 2^2)$. Find $P(|E| > 3)$.

Solution:

$$P(|E| > 3) = 2P(Z > 1.5) = 2(1 - \Phi(1.5)) = 2(0.0668) = 0.1336.$$

Solved Example Finding the Mean

Rope strength $X \sim N(\mu, 20^2)$. 90% of ropes have strength above 200 kg. Find μ.

Solution: We need $P(X \leq 200) = 0.10$, i.e., $P(Z \leq (200 - \mu)/20) = 0.10$. From tables, the 10th percentile is $z = -1.28$. Therefore $(200 - \mu)/20 = -1.28$, so $\mu = 225.6$ kg.

Solved Example Finding the Standard Deviation

A filling machine fills bottles with $N(\mu, \sigma^2)$ volume. The mean is set to 500 mL. If only 2% of bottles contain less than 490 mL, find σ.

Solution: $P(X < 490) = 0.02$, so $(490 - 500)/\sigma = -2.054$ (the 2nd percentile of standard normal). Thus

$$\sigma = \frac{10}{2.054} \approx 4.868 \text{ mL}.$$

Solved Example Control Limits

A process characteristic $X \sim N(50, 2^2)$. Find the symmetric limits within which 99% of items fall.

Solution: We want $P(\mu - k\sigma < X < \mu + k\sigma) = 0.99$. By symmetry, $P(Z < k) = 0.995$, so $k \approx 2.576$. The limits are

$$50 \pm 2.576 \cdot 2 = 50 \pm 5.152 = [44.848,\ 55.152].$$

Solved Example Sum of Independent Normals

Two resistors have resistances $R_1 \sim N(100, 4)$ and $R_2 \sim N(150, 9)$, independent. Find $P(R_1 + R_2 > 255)$.

Solution: $R_1 + R_2 \sim N(250, 13)$.

$$P(R_1 + R_2 > 255) = P\left(Z > \frac{255 - 250}{\sqrt{13}}\right) = P(Z > 1.387) \approx 0.0827.$$

Solved Example Difference of Independent Normals

If $X \sim N(100, 25)$ and $Y \sim N(90, 16)$ independent, find $P(X - Y > 5)$.

Solution: $X - Y \sim N(10, 41)$.

$$P(X - Y > 5) = P\left(Z > \frac{5 - 10}{\sqrt{41}}\right) = P(Z > -0.781) = 0.7823.$$

Solved Example Sample Mean

If X_1, X_2, X_3 are i.i.d. $N(10, 4)$, find the distribution of the sample mean $\bar{X} = \frac{X_1 + X_2 + X_3}{3}$.

Solution: Each X_i has mean 10 and variance 4. The sum $S = X_1 + X_2 + X_3$ has mean 30 and variance 12, so $\bar{X} = S/3$ has

$$E[\bar{X}] = 10, \quad \text{Var}(\bar{X}) = \frac{12}{9} = \frac{4}{3}.$$

Thus $\bar{X} \sim N(10, 4/3)$.

Solved Example Normal Approximation to the Binomial

A fair coin is tossed 100 times. Estimate $P(45 \leq X \leq 55)$, where X is the number of heads, using the normal approximation.

Solution: $X \sim B(100, 0.5)$, so $\mu = 50$, $\sigma = \sqrt{100 \cdot 0.5 \cdot 0.5} = 5$. With continuity correction:

$$P(44.5 \leq X \leq 55.5) \approx P\left(\frac{44.5 - 50}{5} \leq Z \leq \frac{55.5 - 50}{5}\right) = P(-1.1 \leq Z \leq 1.1).$$

$$= 2\Phi(1.1) - 1 = 2(0.8643) - 1 = 0.7286.$$

(The exact binomial value is about 0.7287, so the approximation is excellent.)

8 Transformation of Continuous Random Variables

8.1 The General Problem

Given a random variable X with known PDF $f_X(x)$, and a function $Y = g(X)$, how do we find the PDF $f_Y(y)$ of Y?

There are two main methods:

1. **Distribution function (CDF) method**: find $F_Y(y) = P(Y \leq y)$ and differentiate.
2. **Jacobian (change-of-variable) method**: use the change-of-variable formula when g is one-to-one.

8.2 Method 1: Distribution Function Method

This method always works, regardless of whether g is monotonic:

1. Find $F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$, expressed as a probability involving X.
2. Differentiate $F_Y(y)$ to get $f_Y(y) = \frac{d}{dy} F_Y(y)$.

The CDF method is essential when the transformation is not one-to-one (e.g., $Y = X^2$, $Y = |X|$, $Y = \sin X$).

8.3 Method 2: Jacobian (Change-of-Variable) Method

If $Y = g(X)$ and g is one-to-one and differentiable (strictly monotonic), with inverse $x = g^{-1}(y)$, then

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left|\frac{dx}{dy}\right|}$$

where $\frac{dx}{dy}$ is the derivative of the inverse function.

The absolute value is needed because if g is decreasing, $dx/dy < 0$, but the PDF must be non-negative.

8.4 How to Choose a Method

- If g is strictly increasing or strictly decreasing: the Jacobian method is usually faster.
- If g is not monotonic (e.g., a quadratic, absolute value, trigonometric): use the CDF method.
- If g is linear: either method gives the same answer, but the Jacobian method is immediate.

8.5 Linear Transformation

If $Y = aX + b$ with $a \neq 0$, then

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right).$$

This is one of the most useful transformation formulas and is used constantly in standardisation.

Solved Example Standardising a Normal

If $X \sim N(\mu, \sigma^2)$ and $Z = (X-\mu)/\sigma$, show that $Z \sim N(0,1)$.

Solution: Here $a = 1/\sigma$, $b = -\mu/\sigma$, so $|a| = 1/\sigma$ and $x = \sigma z + \mu$. Then

$$f_Z(z) = \sigma \cdot f_X(\sigma z + \mu) = \sigma \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(\sigma z+\mu-\mu)^2}{2\sigma^2}} = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

This is the standard normal density.

Solved Example Exponential Linear Transformation

If $X \sim \text{Exp}(\lambda)$ and $Y = aX$ with $a > 0$, find f_Y.

Solution: $x = y/a$, $dx/dy = 1/a$.

$$f_Y(y) = f_X(y/a) \cdot \frac{1}{a} = \lambda e^{-\lambda y/a} \cdot \frac{1}{a} = \frac{\lambda}{a} e^{-(\lambda/a)y}, \quad y > 0.$$

So $Y \sim \text{Exp}(\lambda/a)$: scaling an exponential changes only its rate.

Solved Example Uniform Linear Transformation

If $U \sim U(0,1)$ and $Y = a + (b-a)U$, find f_Y.

Solution: $u = (y-a)/(b-a)$, $du/dy = 1/(b-a)$.

$$f_Y(y) = 1 \cdot \frac{1}{b-a}, \quad a < y < b.$$

So $Y \sim U(a,b)$. This is the inverse-transform method for generating a uniform on any interval.

8.6 Power Transformations

Solved Example Square Root of a Uniform (Jacobian Method)

If $X \sim U(0,1)$ and $Y = \sqrt{X}$, find f_Y.

Solution: The transformation is one-to-one on the support of X. Inverse: $x = y^2$, $dx/dy = 2y$.

$$f_Y(y) = f_X(y^2) \cdot 2y = 1 \cdot 2y = 2y, \quad 0 < y < 1.$$

Check: $\int_0^1 2y\,dy = 1$. ✓

Solved Example Square of an Exponential

If $X \sim \text{Exp}(1)$ and $Y = X^2$, find f_Y.

Solution: Since $X > 0$, the transformation is monotonic. Inverse: $x = \sqrt{y}$, $dx/dy = 1/(2\sqrt{y})$.

$$f_Y(y) = f_X(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} = e^{-\sqrt{y}} \cdot \frac{1}{2\sqrt{y}} = \frac{e^{-\sqrt{y}}}{2\sqrt{y}}, \quad y > 0.$$

Solved Example Log of a Uniform

If $X \sim U(0,1)$ and $Y = -\ln X$, find f_Y.

Solution: $x = e^{-y}$, $dx/dy = -e^{-y}$, $|dx/dy| = e^{-y}$.

$$f_Y(y) = 1 \cdot e^{-y} = e^{-y}, \quad y > 0.$$

So $Y \sim \text{Exp}(1)$. This gives the inverse-transform method for generating exponential random variables.

Solved Example Log of a General Uniform

If $X \sim U(0,1)$ and $Y = -\frac{1}{\lambda}\ln X$, find f_Y.

Solution: $x = e^{-\lambda y}$, $dx/dy = -\lambda e^{-\lambda y}$, $|dx/dy| = \lambda e^{-\lambda y}$.

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y > 0.$$

So $Y \sim \text{Exp}(\lambda)$.

Solved Example Inverse of a Uniform

If $X \sim U(0,1)$ and $Y = 1/X$, find f_Y.

Solution: The transformation is one-to-one for $X \in (0,1)$, and the range is $Y \in (1,\infty)$. Inverse: $x = 1/y$, $dx/dy = -1/y^2$, $|dx/dy| = 1/y^2$.

$$f_Y(y) = 1 \cdot \frac{1}{y^2} = \frac{1}{y^2}, \quad y > 1.$$

This is a Pareto distribution with shape 1.

Solved Example Exponential of a Normal

If $X \sim N(0,1)$ and $Y = e^X$, find f_Y. What distribution is this?

Solution: The transformation is one-to-one. Inverse: $x = \ln y$, $dx/dy = 1/y$.

$$f_Y(y) = f_X(\ln y) \cdot \frac{1}{y} = \frac{1}{y\sqrt{2\pi}} e^{-(\ln y)^2/2}, \quad y > 0.$$

This is the **lognormal** distribution with $\mu = 0$, $\sigma = 1$.

8.7 Multivariable Transformations (Brief)

If (X, Y) is a bivariate random vector with joint density $f_{X,Y}(x, y)$, and $(U, V) = (g_1(X, Y), g_2(X, Y))$ is a one-to-one transformation with inverse $(x, y) = (h_1(u, v), h_2(u, v))$, then the joint density of (U, V) is

$$f_{U,V}(u, v) = f_{X,Y}(h_1(u, v), h_2(u, v)) \cdot |J|,$$

where J is the Jacobian determinant

$$J = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}.$$

This extends the single-variable Jacobian formula and is used extensively in statistics (e.g., deriving the distribution of X/Y or $X + Y$).

9 Transformation of Discrete Random Variables

9.1 Introduction

In the continuous case, transforming $Y = g(X)$ requires integration and differentiation (the CDF method) or the Jacobian formula. In the discrete case, there is no integration and no Jacobian. The transformation is handled by **collecting probabilities**. When the transformation is **one-to-one**, the rule is especially simple: each value of y corresponds to exactly one value of x, and the probability is simply carried over.

This section focuses on **one-to-one transformations** of discrete random variables. Non-one-to-one transformations (such as $Y = X^2$ on a symmetric support, $Y = |X|$, or $Y = \max(X_1, X_2)$) require summing probabilities over pre-images and are treated separately.

9.2 The One-to-One Transformation Rule

Let X be a discrete random variable with PMF $p_X(x)$ and support $\mathcal{S}_X$. Let $Y = g(X)$, where g is a **one-to-one** function on $\mathcal{S}_X$. Then for each y in the range of g,

$$\boxed{p_Y(y) = P(Y = y) = P(g(X) = y) = P(X = g^{-1}(y)) = p_X\big(g^{-1}(y)\big).}$$

If y is not in the range of g, then $p_Y(y) = 0$.

Key points.

1. Because g is one-to-one, the probability mass at x is moved entirely to $y = g(x)$.
2. The support of Y is $\mathcal{S}_Y = \{g(x) : x \in \mathcal{S}_X\}$, which has the same number of points as $\mathcal{S}_X$.
3. There is no Jacobian and no absolute derivative in the discrete case.

9.3 Linear Transformations

For $Y = aX + b$ with $a \neq 0$, the map is one-to-one. Hence

$$\boxed{p_Y(y) = p_X\left(\frac{y-b}{a}\right), \qquad y \in \{ax + b : x \in \mathcal{S}_X\}.}$$

The mean and variance follow the standard rules:

$$E[Y] = aE[X] + b, \qquad \mathrm{Var}(Y) = a^2\,\mathrm{Var}(X).$$

For the moment generating function,

$$M_Y(t) = E[e^{t(aX+b)}] = e^{bt}M_X(at).$$

9.4 Examples of One-to-One Transformations

Solved Example Linear Transformation of a Fair Die

A fair die is rolled and X is the face value, so $p_X(x) = 1/6$ for $x = 1, 2, 3, 4, 5, 6$. Let $Y = 2X + 1$. Find the PMF of Y, and verify $E[Y] = 2E[X] + 1$.

Solution: The map $g(x) = 2x + 1$ is one-to-one. The range is

$$y \in \{3, 5, 7, 9, 11, 13\}.$$

The inverse is $x = (y - 1)/2$. Therefore

$$p_Y(y) = p_X\left(\frac{y-1}{2}\right) = \frac{1}{6}, \qquad y \in \{3, 5, 7, 9, 11, 13\}.$$

Check the sum: $6 \times \frac{1}{6} = 1$. ✓

Mean:

$$E[Y] = 2E[X] + 1 = 2(3.5) + 1 = 8.$$

Directly:

$$E[Y] = \frac{3 + 5 + 7 + 9 + 11 + 13}{6} = \frac{48}{6} = 8.\checkmark$$

Variance:

$$\mathrm{Var}(Y) = 4\,\mathrm{Var}(X) = 4 \cdot \frac{35}{12} = \frac{35}{3} \approx 11.6667.$$

Solved Example Complement Transformation of a Binomial

Let $X \sim \mathrm{Binomial}(n, p)$, so

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n.$$

Define $Y = n - X$, the number of failures. Find the PMF of Y.

Solution: The map $g(x) = n-x$ is one-to-one on $\{0, 1, \ldots, n\}$. The inverse is $x = n-y$. Therefore

$$p_Y(y) = p_X(n-y) = \binom{n}{n-y} p^{n-y}(1-p)^y = \binom{n}{y}(1-p)^y p^{n-y}, \qquad y = 0, 1, \ldots, n.$$

This is the Binomial$(n, 1-p)$ PMF. So the number of failures follows a binomial distribution with success probability $1-p$, which is intuitive: failures are "successes" when the roles of success and failure are swapped.

Mean and variance:

$$E[Y] = n - E[X] = n - np = n(1-p), \qquad \mathrm{Var}(Y) = \mathrm{Var}(X) = np(1-p).$$

Solved Example Scaling a Poisson

Let $X \sim \text{Poisson}(\lambda)$, so $p_X(x) = e^{-\lambda}\lambda^x/x!$ for $x = 0, 1, 2, \ldots$. Define $Y = 2X$. Find the PMF of Y, and compute $E[Y]$ and $\mathrm{Var}(Y)$.

Solution: The map $g(x) = 2x$ is one-to-one on the nonnegative integers: it maps x to the even nonnegative integer $2x$. The inverse is $x = y/2$, which is defined only when y is even. Thus

$$p_Y(y) = \begin{cases} p_X\left(\dfrac{y}{2}\right) = \dfrac{e^{-\lambda}\lambda^{y/2}}{(y/2)!}, & y = 0, 2, 4, 6, \ldots, \\ 0, & \text{otherwise.} \end{cases}$$

This is a *scaled Poisson* distribution supported on the even nonnegative integers.

Mean and variance:

$$E[Y] = 2E[X] = 2\lambda, \qquad \mathrm{Var}(Y) = 4\,\mathrm{Var}(X) = 4\lambda.$$

Solved Example Shift of a Geometric

Let $X \sim \text{Geometric}(p)$, so

$$p_X(x) = (1-p)^{x-1}p, \qquad x = 1, 2, 3, \ldots$$

Define $Y = X + 5$. Find the PMF of Y, and compute $E[Y]$.

Solution: The map $g(x) = x + 5$ is one-to-one. The inverse is $x = y - 5$, with $y \in \{6, 7, 8, \ldots\}$. Hence

$$p_Y(y) = p_X(y-5) = (1-p)^{(y-5)-1}p = (1-p)^{y-6}p, \qquad y = 6, 7, 8, \ldots$$

This is a shifted geometric distribution starting at 6 instead of 1.

Mean:

$$E[Y] = E[X] + 5 = \frac{1}{p} + 5.$$

Variance:

$$\mathrm{Var}(Y) = \mathrm{Var}(X) = \frac{1-p}{p^2}.$$

Solved Example Reciprocal Transformation of a Discrete Uniform

Let X be uniformly distributed on $\{1,2,3,4,5,6\}$, so $p_X(x) = 1/6$ for each x. Define $Y = 1/X$. Find the PMF of Y.

Solution: The map $g(x) = 1/x$ is one-to-one on the positive support $\{1,\ldots,6\}$. The range is

$$y \in \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}\right\}.$$

The inverse is $x = 1/y$. Therefore

$$p_Y(y) = p_X\left(\frac{1}{y}\right) = \frac{1}{6}, \qquad y \in \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}\right\}.$$

The distribution remains uniform on the transformed support, because the transformation is one-to-one and the original distribution is uniform.

Mean (not equal to $1/E[X]$ in general):

$$E[Y] = \frac{1}{6}\left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}\right) = \frac{1}{6} \cdot \frac{49}{20} \approx 0.4083.$$

Note that $1/E[X] = 1/3.5 \approx 0.2857$, so $E[1/X] \neq 1/E[X]$. This is an important warning: nonlinear transformations do not commute with expectation.

Solved Example Quadratic Transformation of a Poisson

Let $X \sim \mathrm{Poisson}(\lambda)$ on $\{0,1,2,\ldots\}$. Define $Y = X^2$. Show that the map is one-to-one on the support, and find the PMF of Y.

Solution: The map $g(x) = x^2$ is one-to-one on the nonnegative integers: distinct nonnegative integers have distinct squares ($0^2 = 0, 1^2 = 1, 2^2 = 4, 3^2 = 9, \ldots$). The inverse is $x = \sqrt{y}$, which is defined only when y is a perfect square. Thus

$$p_Y(y) = \begin{cases} p_X(\sqrt{y}) = \dfrac{e^{-\lambda}\lambda^{\sqrt{y}}}{(\sqrt{y})!}, & y \in \{0,1,4,9,16,\ldots\}, \\ 0, & \text{otherwise.} \end{cases}$$

The support of Y is the set of perfect squares, which is a subset of the nonnegative integers. The transformation is one-to-one on the original support, but the new support is sparse.

Mean:

$$E[Y] = E[X^2] = \mathrm{Var}(X) + (E[X])^2 = \lambda + \lambda^2.$$

10 Summary Table of Continuous Distributions

Table 1: Summary of continuous distributions

Distribution	PDF	Mean	Variance
Uniform(a,b)	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential(λ)	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gamma(α,β)	$\frac{x^{\alpha-1}e^{-x/\beta}}{\Gamma(\alpha)\beta^\alpha}$	$\alpha\beta$	$\alpha\beta^2$
Beta(α,β)	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Normal(μ,σ^2)	$\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2

11 Relationships Between the Distributions

- Uniform$(0,1)$ = Beta$(1,1)$.
- Exponential(λ) = Gamma$(1,1/\lambda)$.
- Gamma$(n/2,2) = \chi^2_n$.
- If $X \sim N(0,1)$, then $X^2 \sim \chi^2_1$ = Gamma$(1/2,2)$.
- If $U \sim U(0,1)$, then $-\frac{1}{\lambda}\ln U \sim$ Exp(λ).

12 Practice Problems

1. Evaluate $\Gamma(6)$, $\Gamma(9/2)$, and $\displaystyle\int_0^\infty x^4 e^{-3x}\,dx$.
2. $X \sim U(-1,3)$. Find $E[X]$, $\mathrm{Var}(X)$, $P(X>1)$, $P(|X|<1)$.
3. Lifetime is exponential with mean 800 hours. Find $P(T>1000)$, the median, and the hazard rate.
4. Time until 5th arrival in a Poisson process with rate 3/hour. Find the mean and variance.
5. $X \sim$ Beta$(2,5)$. Find the mean, variance, and $P(X<0.2)$.
6. $X \sim$ Beta$(2,2)$. Find the mode, $E[X]$, and $P(X<0.5)$.
7. A proportion follows Beta$(3,7)$. Find the probability that the proportion exceeds 0.5.
8. Quality characteristic $N(50,2^2)$. Find $P(|X-50|<3)$ and the symmetric limits containing 95% of items.
9. If $X \sim N(10,4)$, find $P(8<X<13)$ and $P(X>15)$.
10. $X \sim U(0,1)$, $Y=-2\ln X$. Find $f_Y(y)$.
11. $X \sim$ Exp(1), $Y=3X+2$. Find $f_Y(y)$.
12. $X \sim$ Gamma$(3,2)$. Find $P(X \le 10)$ using the CDF formula for integer α.

13. Two independent components have lifetimes $\text{Exp}(\lambda_1 = 0.01)$ and $\text{Exp}(\lambda_2 = 0.02)$. Find the probability that the first one outlasts the second.

14. Using the normal approximation, estimate $P(48 \le X \le 52)$ for $X \sim B(100, 0.5)$, with and without continuity correction.

15. A random variable X is uniformly distributed on $\{-2, -1, 0, 1, 2\}$, so $p_X(x) = 1/5$ for each x. Let $Y = 3X - 4$.

    (a) Find the PMF of Y.
    (b) Compute $E[Y]$ and $\text{Var}(Y)$ using the linear transformation rules, and verify directly.

16. Let $X \sim \text{Binomial}(4, 0.3)$. Define $Y = X + 2$.

    (a) Find the PMF of Y.
    (b) Compute $P(Y \le 3)$.
    (c) Compute $E[Y]$ and $\text{Var}(Y)$.

17. Let $X \sim \text{Poisson}(3)$. Define $Y = 2X + 1$.

    (a) Find the PMF of Y.
    (b) Compute $P(Y = 5)$.
    (c) Compute $E[Y]$ and $\text{Var}(Y)$.

18. Let X be uniformly distributed on $\{1, 2, 3, 4, 5\}$. Define $Y = 6 - X$.

    (a) Show that Y is also uniformly distributed on $\{1, 2, 3, 4, 5\}$.
    (b) Compute $E[Y]$ and $\text{Var}(Y)$.
    (c) Is the distribution of Y the same as the distribution of X? Explain.

19. Let $X \sim \text{Geometric}(p)$ with $p = 0.25$. Define $Y = X + 3$.

    (a) Find the PMF of Y.
    (b) Compute $P(Y \le 5)$.
    (c) Compute $E[Y]$ and $\text{Var}(Y)$.

20. Let X be uniformly distributed on $\{1, 2, 3, 4\}$. Define $Y = 2^X$.

    (a) Show that Y is one-to-one on the support.
    (b) Find the PMF of Y.
    (c) Compute $E[Y]$.

13 Solution

15. (a) The map $y = 3x - 4$ is one-to-one. The support of X is $\{-2, -1, 0, 1, 2\}$, so the support of Y is $\{-10, -7, -4, -1, 2\}$. Since $p_X(x) = 1/5$ for each x, $p_Y(y) = 1/5$ for each y in that set.

    (b) $E[Y] = 3E[X] - 4 = 3(0) - 4 = -4$. Directly: $(-10 - 7 - 4 - 1 + 2)/5 = -20/5 = -4$. ✓

$$\text{Var}(Y) = 9\,\text{Var}(X) = 9 \cdot \frac{4 + 1 + 0 + 1 + 4}{5} = 9 \cdot 2 = 18.$$

    Direct check: $E[Y^2] = (100 + 49 + 16 + 1 + 4)/5 = 170/5 = 34$, so $\text{Var}(Y) = 34 - 16 = 18$. ✓

16. (a) $p_X(0) = 0.2401$, $p_X(1) = 0.4116$, $p_X(2) = 0.2646$, $p_X(3) = 0.0756$, $p_X(4) = 0.0081$. The support of $Y = X + 2$ is $\{2, 3, 4, 5, 6\}$, with

$$p_Y(2) = 0.2401,\ p_Y(3) = 0.4116,\ p_Y(4) = 0.2646,\ p_Y(5) = 0.0756,\ p_Y(6) = 0.0081.$$

(b) $P(Y \le 3) = P(X \le 1) = 0.2401 + 0.4116 = 0.6517$.
(c) $E[Y] = E[X] + 2 = 1.2 + 2 = 3.2$. $\mathrm{Var}(Y) = \mathrm{Var}(X) = 4(0.3)(0.7) = 0.84$.

17. (a) $p_Y(y) = p_X((y-1)/2) = e^{-3}3^{(y-1)/2}/(((y-1)/2)!$ for $y = 1, 3, 5, 7, \ldots$ (odd positive integers); zero otherwise.
(b) $P(Y = 5) = P(X = 2) = e^{-3}3^2/2! = 9e^{-3}/2 \approx 0.2240$.
(c) $E[Y] = 2E[X] + 1 = 6 + 1 = 7$. $\mathrm{Var}(Y) = 4\,\mathrm{Var}(X) = 4(3) = 12$.

18. (a) $Y = 6 - X$ maps $\{1, 2, 3, 4, 5\}$ onto $\{1, 2, 3, 4, 5\}$ bijectively. For each y, $p_Y(y) = p_X(6 - y) = 1/5$, so Y is uniform on the same set.
(b) $E[Y] = 3$, $\mathrm{Var}(Y) = 2$ (same as X, since the distribution is identical).
(c) Yes, the distribution is the same, even though the transformation is not the identity.

19. (a) $p_Y(y) = p_X(y - 3) = (0.75)^{y-4}(0.25)$, $y = 4, 5, 6, \ldots$.
(b) $P(Y \le 5) = P(X \le 2) = 1 - (1 - p)^2 = 1 - 0.75^2 = 1 - 0.5625 = 0.4375$.
(c) $E[Y] = 1/0.25 + 3 = 4 + 3 = 7$. $\mathrm{Var}(Y) = (1 - 0.25)/(0.25)^2 = 0.75/0.0625 = 12$.

20. (a) $g(x) = 2^x$ is strictly increasing, hence one-to-one, on $\{1, 2, 3, 4\}$.
(b) Support of Y is $\{2, 4, 8, 16\}$, each with probability $1/4$.
(c) $E[Y] = (2 + 4 + 8 + 16)/4 = 30/4 = 7.5$.